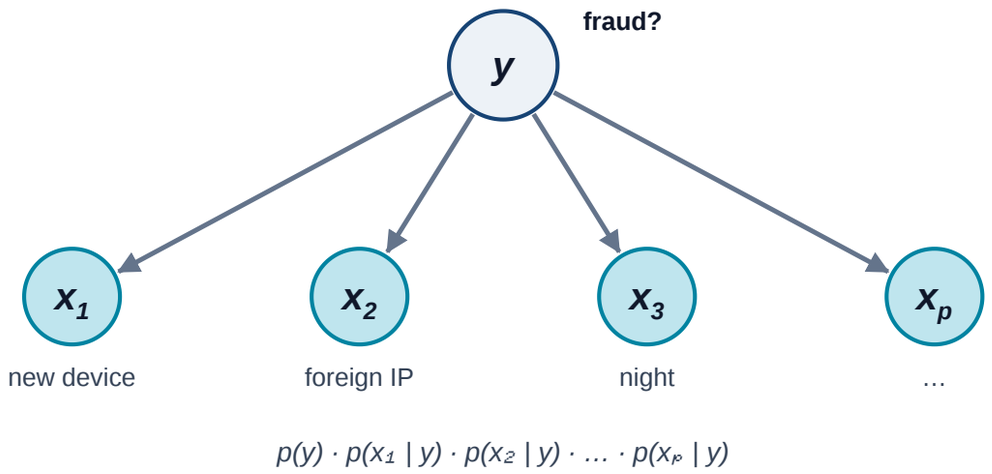
Probabilistic Machine Learning

Lecture 05: Bayesian Networks

Prof. Dr. Michael Granitzer



Where We Left Off



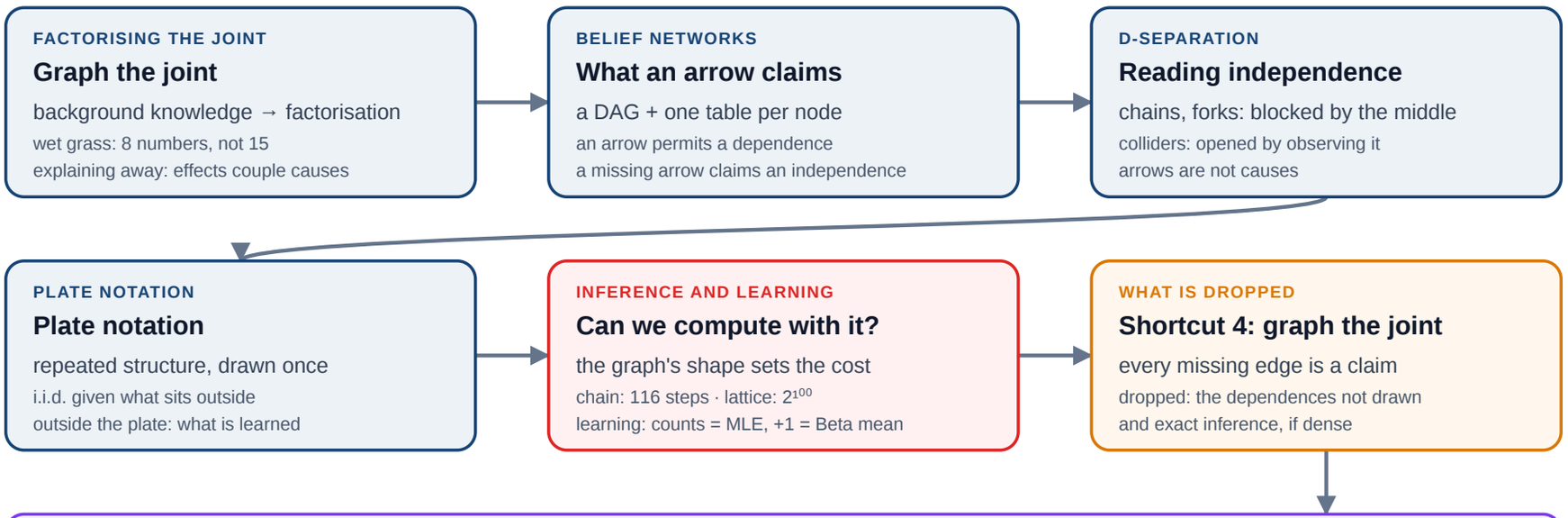
Lecture 04 skipped $p(x)$. Lecture 03 kept the whole joint and made it affordable with one blanket assumption: **every** attribute independent of every other, given the class. It worked — and it counted correlated evidence twice.

What if we assumed only the independences we actually believe?

Shortcut 4 — graph the joint. Draw which variables depend on which; the graph *is* the factorisation. Dropped: every dependence **not drawn** — and, for a dense graph, affordable inference.

Argument Map

Lecture 05 · argument map



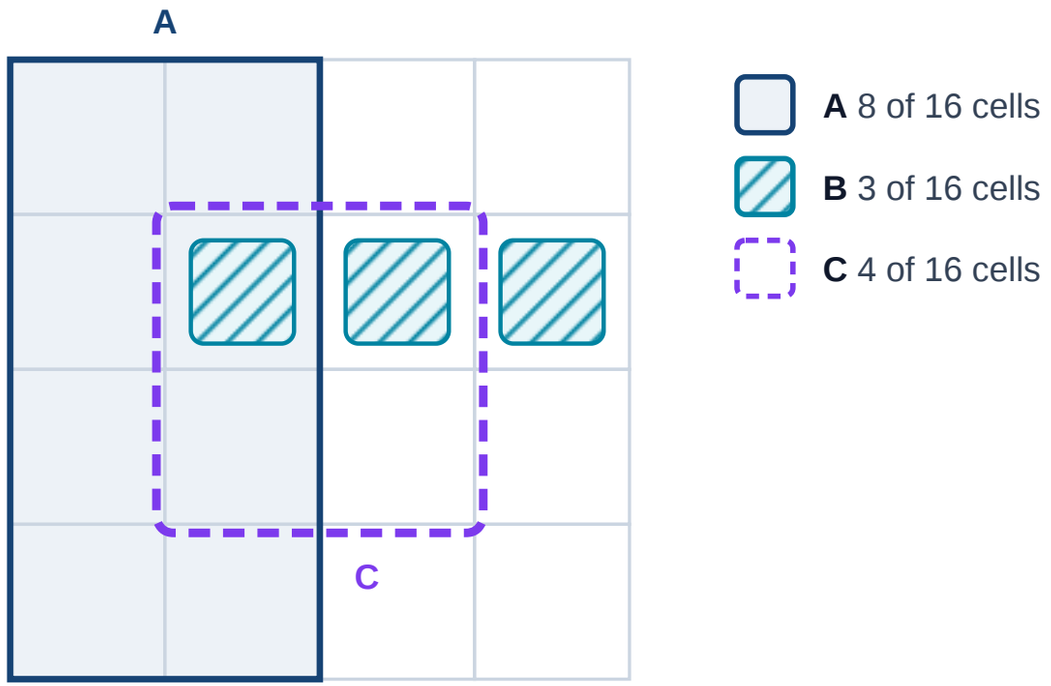
Course arc
ideal → shortcuts

L01	L02	L03	L04	L05	L06	L07
The ideal	The language	Apart · peak	Only p(y | x)	Graph the joint	Keep the loss	No shortcut

Factorising the Joint

Guiding Question: Naive Bayes assumed every attribute independent of every other. What if we stated only the independences we believe — and what does conditioning then do to them?

Independent Given C — Counted



Sixteen equally likely cells.

Dependent

$$P(A \cap B) = \frac{1}{16} \neq \frac{3}{32} = P(A)P(B)$$

Independent given C

$$P(A \cap B \mid C) = \frac{1}{4} = P(A \mid C)P(B \mid C)$$

Why Is the Grass Wet?

Tracey sees her lawn is wet. **Rain — or did she leave the sprinkler on?** Then she sees that her neighbour Jack's lawn is wet too.

variable	meaning (1 = yes)
R	it rained
S	Tracey's sprinkler was on
J	Jack's grass is wet
T	Tracey's grass is wet

Before any arithmetic: does Jack's wet lawn make the sprinkler **more** or **less** likely?

Background Knowledge as Independence

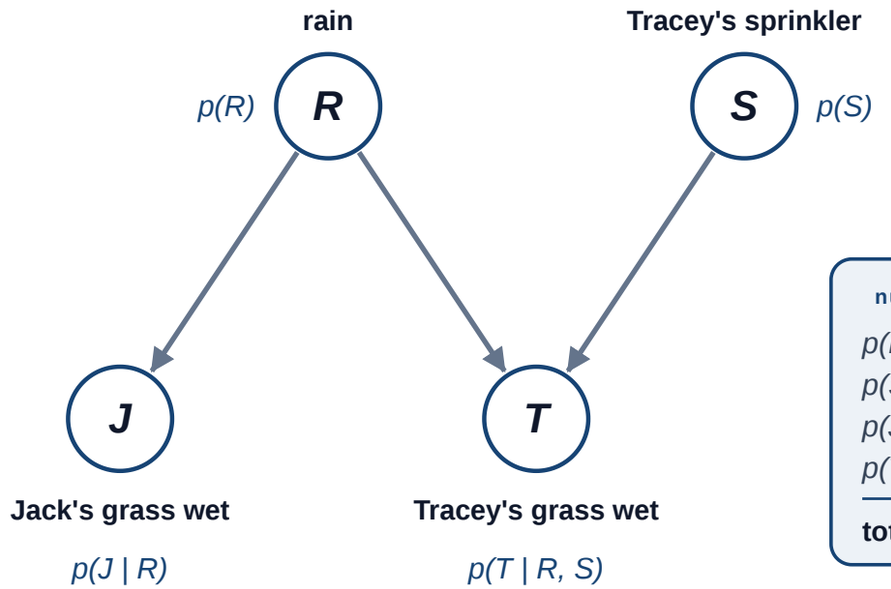
The joint needs $2^4 - 1 = 15$ numbers. The chain rule rewrites it without shrinking it:

$$p(T, J, R, S) = p(T \mid J, R, S) \cdot p(J \mid R, S) \cdot p(R \mid S) \cdot p(S)$$

- → Tracey's lawn ignores Jack's, given R and S: $p(T \mid J, R, S) = p(T \mid R, S)$
- → Jack's lawn ignores Tracey's sprinkler: $p(J \mid R, S) = p(J \mid R)$
- → the sprinkler does not change the weather: $p(R \mid S) = p(R)$

> $$p(T, J, R, S) = p(T \mid R, S) \cdot p(J \mid R) \cdot p(R) \cdot p(S)$$

The Graph Is the Factorisation



numbers needed	
p(R)	**1**
p(S)	**1**
p(J | R)	**2**
p(T | R, S)	**4**
total	**8**

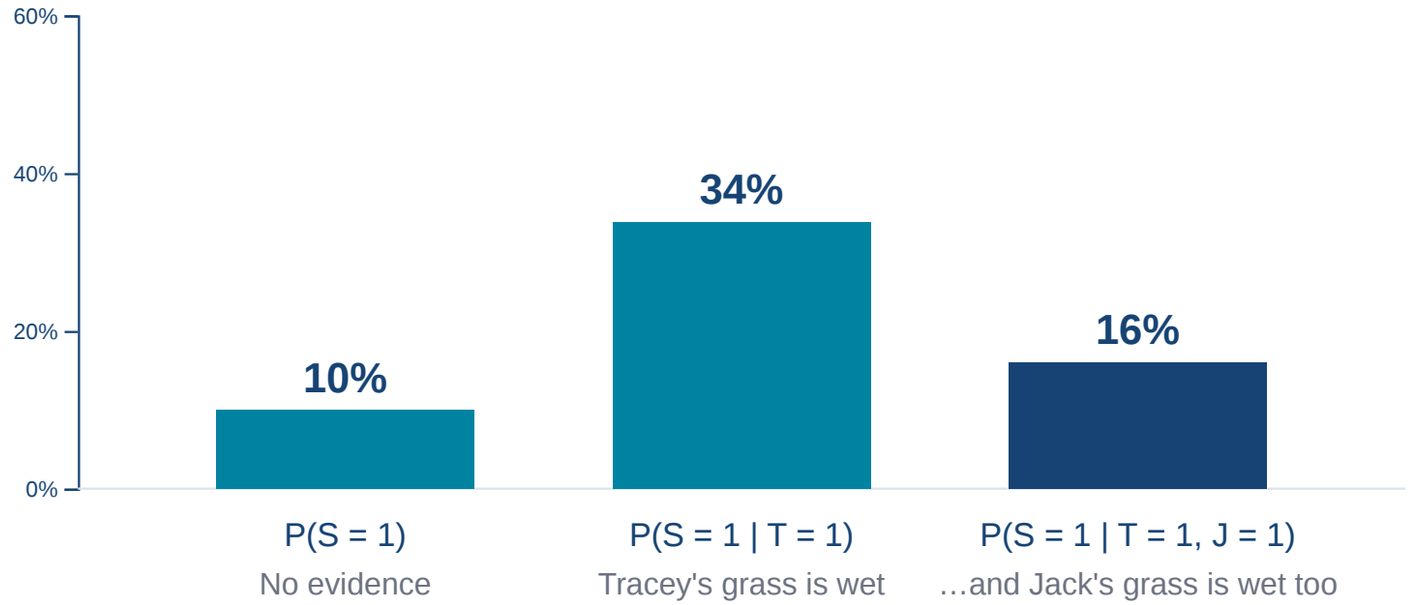
full joint: 15

factor	$= 1$
$p(R = 1)$	0.2
$p(S = 1)$	0.1
$p(J = 1 \mid R = 0)$	0.2
$p(J = 1 \mid R = 1)$	1.0
$p(T = 1 \mid R = 0, S = 0)$	0.0
$p(T = 1 \mid R = 0, S = 1)$	0.9
$p(T = 1 \mid R = 1, S = 0)$	1.0
$p(T = 1 \mid R = 1, S = 1)$	1.0

One node per variable, one arrow per conditioning bar, one table per node.

What Does the Evidence Say?

Rain explains the wetness — the sprinkler falls back towards its prior



Explaining Away

$$p(S=1 \mid T=1, J=1) = \frac{\sum_R p(T{=}1 \mid R, S{=}1)p(J{=}1 \mid R)p(R)p(S{=}1)}{\sum_{R,S} p(T{=}1 \mid R, S)p(J{=}1 \mid R)p(R)p(S)} = \frac{0.0344}{0.2144} \approx 0.16$$

Rain and sprinkler are independent: $R \perp\!\!\!\perp S$. Once their **common effect** T is observed they compete to explain it, and evidence for one lowers the other. Conditioning on a common cause removes a dependence; conditioning on a common effect **creates** one.

Mini-Exercise

Tracey's lawn is wet. Then she finds the **sprinkler was on**.

Does rain become more or less likely than it was a moment ago?

Mini-Exercise

Tracey's lawn is wet. Then she finds the **sprinkler was on**.

Does rain become more or less likely than it was a moment ago?

Less. The sprinkler explains the wet lawn. $P(R = 1 \mid T = 1) \approx 0.74$ falls to $P(R = 1 \mid T = 1, S = 1) = 0.02/0.092 \approx 0.22$ — almost back to the prior 0.2. Explaining away works in both directions.

Takeaway

Graph the joint: background knowledge becomes conditional independence, and the graph is a picture of how the joint splits — 8 numbers instead of 15. Every *missing* edge is a claim; conditioning can make variables independent (common cause) or dependent (common effect).

Belief Networks

Guiding Question: What exactly does an arrow in a probabilistic model claim — and what does a *missing* arrow claim?

Belief Networks

Bayesian belief network. A **directed acyclic graph** whose nodes are random variables, with one conditional table per node given its parents:

$$p(x_1, \ldots, x_n) = \prod_{i=1}^{n} p\big(x_i \mid \mathrm{pa}(x_i)\big)$$

An arrow permits a dependence. A missing arrow claims an independence: each variable is independent of its non-descendants, given its parents (the **local Markov property**).

Which independence does the wet-grass graph claim for Jack's lawn J?

Belief Networks

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Which independence does the wet-grass graph claim for Jack's lawn J?

$J \perp\!\!\!\perp \{S, T\} \mid R$: once we know whether it rained, nothing else says more about Jack's lawn.

Counting the Numbers

model, binary variables	full joint	graph
wet grass, 4 nodes	15	$1 + 1 + 2 + 4 = 8$
naive Bayes, class + 30 flags	$2^{31} - 1 \approx 2.1 \cdot 10^9$	$1 + 2 \cdot 30 = 61$
20 nodes, at most 3 parents each	$2^{20} - 1 \approx 10^6$	at most $20 \cdot 2^3 = 160$

Where does the saving come from — and what makes it disappear?

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Where does the saving come from — and what makes it disappear?

A node with k binary parents needs 2^k numbers. The saving comes from **missing arrows**; it disappears as soon as some node has many parents.

Keeping Tables Small: Noisy-OR

A symptom y with many possible causes $x_1, \ldots, x_k$: **each present cause fires on its own** with probability λ_i, and y occurs if any fires:

$$p(y = 0 \mid x_1, \ldots, x_k) = \prod_{i : x_i = 1} (1 - \lambda_i)$$

k numbers instead of 2^k. Two causes present, $\lambda = 0.8$ and 0.6: $p(y = 1) = 1 - 0.2 \cdot 0.4 = 0.92$.

THE ASSUMPTION BOUGHT

Causes act **independently** — no interaction. Two drugs that are harmless alone but dangerous together break it.

Divorcing is the other trick: insert a summary node ("total load") between a group of parents and the child.

Mini-Exercise

Tracey and Jack share a garden hose: add the arrow $S \to J$ to the wet-grass network.

How many numbers does the network need now — and which independence claim is gone?

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Tracey and Jack share a garden hose: add the arrow $S \to J$ to the wet-grass network.

How many numbers does the network need now — and which independence claim is gone?

10. $p(J \mid R, S)$ needs 4 numbers instead of 2: $1 + 1 + 4 + 4$. Gone is $J \perp\!\!\!\perp S \mid R$ — the missing arrow was that claim.

Takeaway

A belief network is **a DAG plus one table per node**. An arrow only permits a dependence; every **missing** arrow claims an independence, and each claim saves numbers. Shrinking tables further — noisy-OR, divorcing — buys more savings with more assumptions, which should be said out loud.

d-Separation

Guiding Question: Two causes are independent until you observe their shared effect. Can we see such things in the picture — before computing a single number?

Three Ways Three Nodes Connect

CHAIN

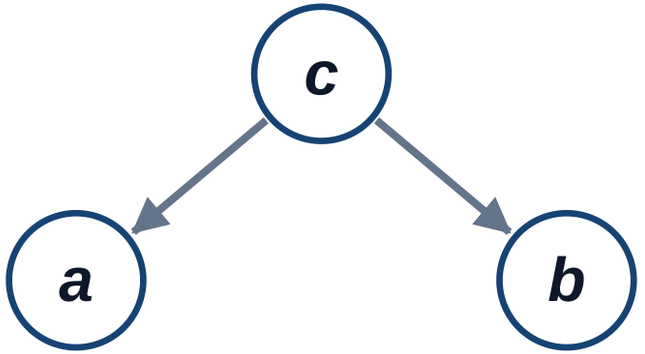


a, b: **dependent**

given c: **independent**

observing c blocks the path

FORK · COMMON CAUSE

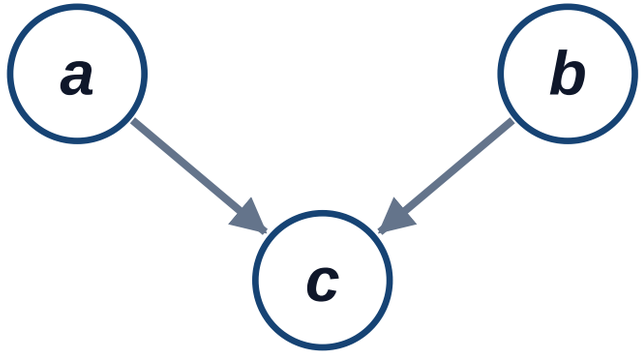


a, b: **dependent**

given c: **independent**

observing c blocks the path

COLLIDER · COMMON EFFECT



a, b: **independent**

given c: **dependent**

observing c opens the path

Every independence question in a DAG reduces to these three.

d-Separation

A path between X and Y is **blocked** by a set Z if it contains

- a chain or fork whose middle node **is in** Z, or
- a collider whose middle node — and every descendant of it — is **not in** Z.

If every path is blocked, X and Y are **d-separated** by Z, and then $X \perp\!\!\!\perp Y \mid Z$ in every distribution that factorises over the graph.

Sound, but not complete for a particular table: specific numbers can create independences the graph does not show. For almost every table that factorises over the graph, d-separation finds them all.

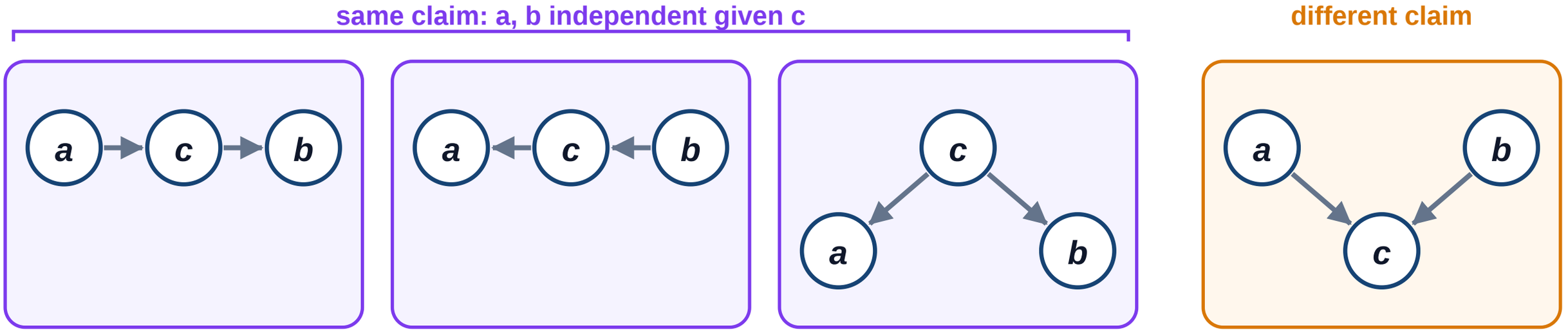
d-Separation on the Wet Grass

The only path from J to S: $J \leftarrow R \rightarrow T \leftarrow S$ — a fork at R, a collider at T.

query	path	verdict	the numbers
$J \perp\!\!\!\perp S$?	collider T unobserved: blocked	independent	$P(J{=}1) = 0.36 = P(J{=}1 \mid S{=}1)$
$J \perp\!\!\!\perp S \mid T$?	collider T observed: open	dependent	$P(J{=}1 \mid T{=}1) = 0.79$, but 0.37 if also $S{=}1$
$J \perp\!\!\!\perp S \mid T, R$?	fork R observed: blocked	independent	$P(S{=}1 \mid T{=}1, R{=}1) = 0.1$, with or without $J{=}1$

> The middle row is the explaining away of Factorising the Joint: $10\% \rightarrow 34\% \rightarrow 16\%$ for the sprinkler. The graph predicted it before any number was computed.

Arrows Are Not Causes



data cannot tell these three apart

the direction of the arrows is not in the data

a, b independent;
dependent given c

Same skeleton, same colliders: **the same independences** (Markov equivalence). Data cannot orient such arrows — they are **not** causal claims.

Mini-Exercise

Tracey's doormat gets wet whenever her lawn is: add $T \rightarrow D$. You see only a **wet doormat —** not the lawn.

Are rain and sprinkler still independent?

Mini-Exercise

Tracey's doormat gets wet whenever her lawn is: add $T \to D$. You see only a **wet doormat** — not the lawn.

Are rain and sprinkler still independent?

No. D is a descendant of the collider T, so observing it opens the path $R \to T \leftarrow S$: a wet doormat also makes rain and sprinkler compete to explain it.

Takeaway

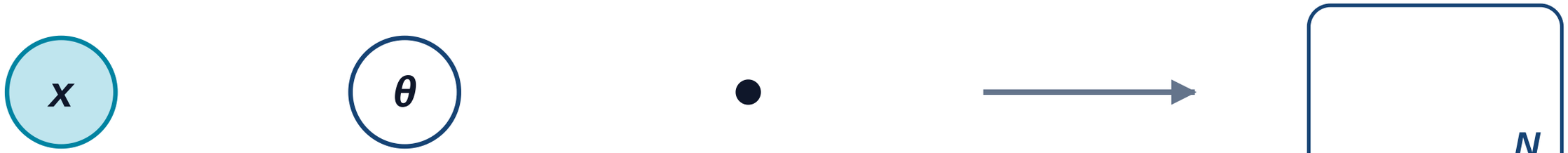
Independence in a belief network is a **property of the graph**: chains and forks are blocked by observing the middle, colliders are opened by it (or by a descendant). d-separation applies that to every path. What it cannot read off is the direction of cause: equivalent graphs make the same claims.

Plate Notation

Guiding Question: A model of 10,000 observations should not need 10,000 nodes. How is a repeated structure drawn — and what does the drawing claim about the observations?

Drawing a Model

A probabilistic model is a factorisation, and the diagram is its picture. Five conventions carry almost all of it:



shaded
observed: in the data

open
latent or parameter

solid dot
fixed, not inferred

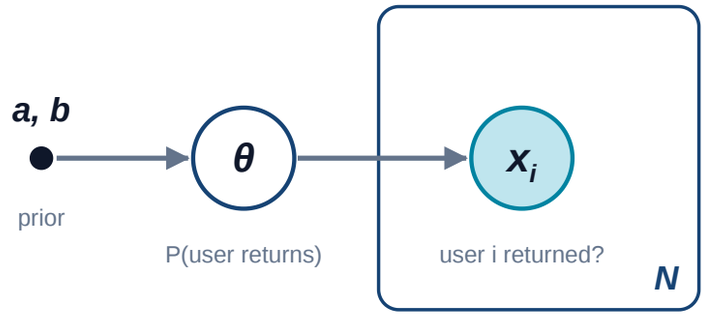
arrow
child conditioned on parent

plate
repeat N times

> Nothing new is claimed: the arrows are read exactly as in a belief network. The shading only says **what the data fixes** — which variables we condition on.

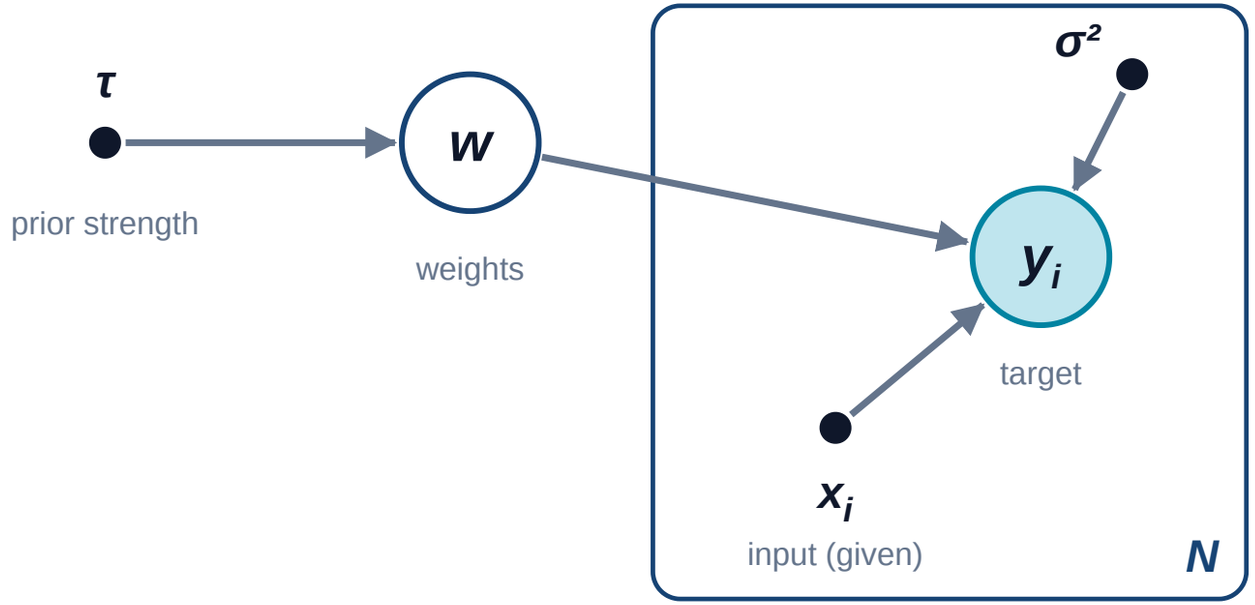
Plates

A **plate** is a box with a count in its corner: *repeat what is inside N times*. A model of 10,000 users does not need 10,000 nodes.
The copies are **independent given what sits outside** — the parameters, which learning estimates. Nested plates draw grouped data: K hospitals with N_k patients each.



> Given θ the users are independent. With θ unknown they are **not**: in Lecture 03, the first ten users changed our prediction for the eleventh. That dependence is exactly why learning from data works.

Reading a Diagram



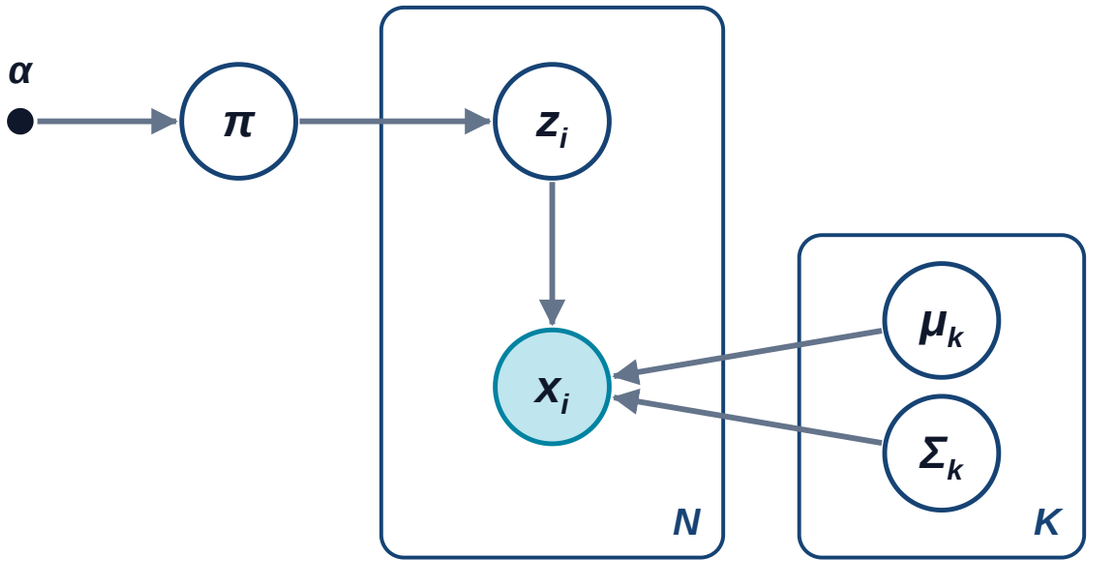
Read off the factorisation:

$$p(w \mid \tau) \prod_{i=1}^{N} p(y_i \mid x_i, w, \sigma^2)$$

The plate became the product; each arrow became a conditioning bar.

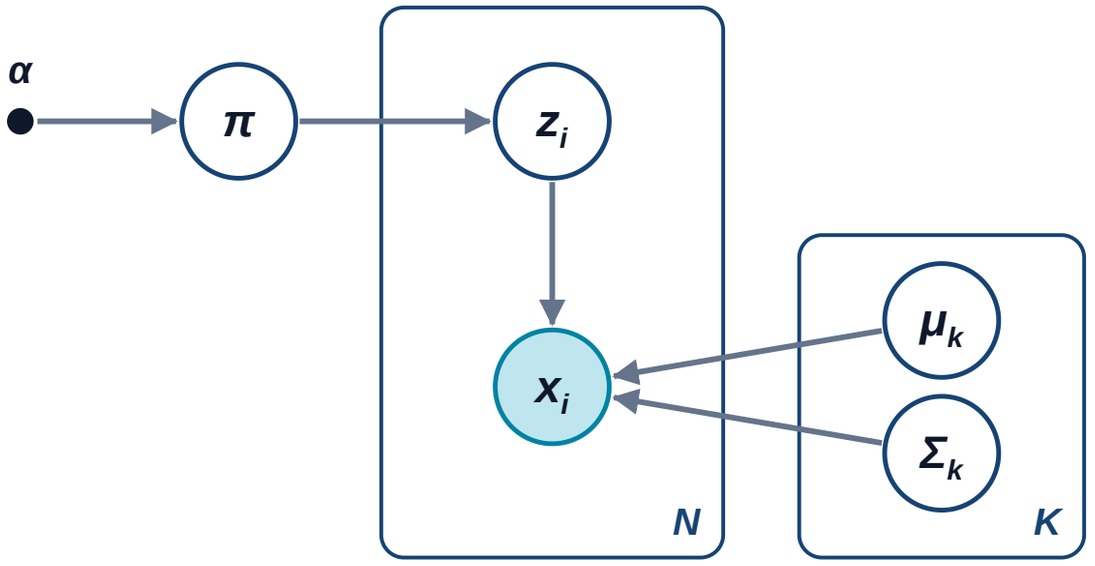
x_i is a **dot**: the inputs are given, not modelled — this is a discriminative model, $p(y \mid x)$, with a prior on w. Lecture 07 conditions on the shaded y_i and keeps the whole posterior over w.

Mini-Exercise



Which nodes does learning have to infer — and why is α not among them?

Mini-Exercise



Which nodes does learning have to infer — and why is α not among them?

π, every μ_k, Σ_k and every label z_i — all **open**.

α is a **dot**: fixed by us. Only the x_i are observed:

$$p(\pi \mid \alpha) \prod_k p(\mu_k, \Sigma_k) \prod_i p(z_i \mid \pi) p(x_i \mid z_i, \mu, \Sigma)$$

A **mixture of Gaussians** — Lecture 07.

Takeaway

A model diagram is a factorisation you can see: **shading** marks what the data fixes, **plates** mark what repeats independently *given* what sits outside, and **what sits outside** is what learning has to estimate.

Inference and Parameter Learning

Guiding Question: We can now write the joint down. Can we actually condition on it — and where do its numbers come from?

Three Questions You Can Ask a Model

question	on the wet grass	with complete data
infer unobserved variables	$P(S \mid T, J)$	cost set by the graph's shape
learn the parameters	the 8 table entries	counting
learn the structure	which arrows at all?	hardest — and only up to equivalence

Inference by Elimination

Explaining away needed $P(T{=}1, J{=}1)$. Sum the factorised joint — and **push each sum as far right as it goes**:

$$P(T{=}1, J{=}1) = \sum_R p(R)p(J{=}1 \mid R) \underbrace{\sum_S p(S)p(T{=}1 \mid R, S)}_{m(R)}$$

Inner sum first

$m(1) = 0.1 \cdot 1 + 0.9 \cdot 1 = 1$

$m(0) = 0.1 \cdot 0.9 + 0.9 \cdot 0 = 0.09$

Then the outer one

$0.2 \cdot 1 \cdot 1 + 0.8 \cdot 0.2 \cdot 0.09$

$= 0.2 + 0.0144 = 0.2144$ ✓

Why Pushing Sums In Pays

A chain of n binary variables $X_1 \to X_2 \to \cdots \to X_n$. To get $p(x_n)$:

$$2^{n-1}$$

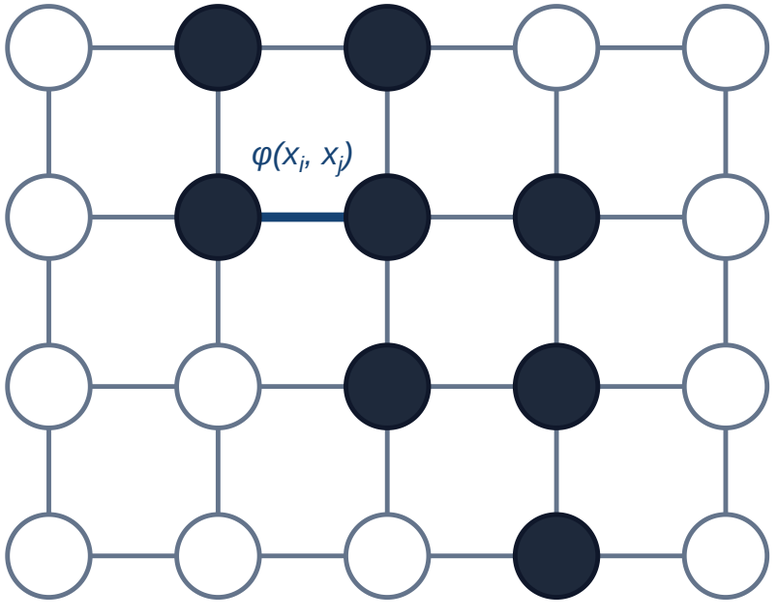
terms if every other variable is summed at once — for $n = 30$, about 537 million

$$4(n-1)$$

operations if the sums are pushed in, one variable at a time — for $n = 30$, 116

Eliminating X_1, then X_2, … passes a **two-entry table** along the chain. The cost is set by the **largest table** created on the way — by the graph's shape, not its size.

Undirected Models



each edge: a potential that prefers neighbours to agree

Pixels influence each other **symmetrically**. A **Markov network** puts a non-negative **potential** ϕ on each edge and normalises once, globally:

$$p(x) = \frac{1}{Z} \prod_{(i,j)} \phi(x_i, x_j), \qquad Z = \sum_{x} \prod_{(i,j)} \phi(x_i, x_j)$$

Two pixels, $\phi = 3$ if equal, 1 if not: $Z = 3 + 1 + 1 + 3 = 8$, so $P(\text{equal}) = 6/8$.

Directed and Undirected

	belief network	Markov network
edges	directed, acyclic	undirected
factors	conditional tables, normalised locally	potentials, one global Z
natural for	causal or generative stories	symmetric interaction: pixels, neighbours
observing a common effect	creates dependence	— (no colliders)

Inference runs on the undirected form: **moralisation** joins parents that share a child (R – S in the wet grass) and drops the arrow heads; elimination works on that graph.

Writable Is Not Queryable

A 100×100 black-and-white image as a pixel lattice:

79,200

numbers to **write it down**: 19,800 edges, 4 per potential

$$2^{100} \approx 10^{30}$$

entries in the table that elimination must build — a whole row of pixels at once

Exact inference in belief networks is **NP-hard** in general, and so is approximate inference with a guaranteed error bound. What remains: exploit the shape where it is kind (chains, trees), and otherwise **sample** or **approximate** — trading the guarantee for an answer.

Learning the Parameters Is Counting

Fifty observed days. Estimate Jack's table $p(J{=}1 \mid R)$:

	days	Jack's lawn wet	MLE (peak)	$+1$ pseudo-counts (mean)
$R=0$	40	8	$8/40 = 0.20$	$9/42 \approx 0.21$
$R=1$	10	10	$10/10 = 1.0$	$11/12 \approx 0.92$

> Counting **is** maximum likelihood (Lecture 03); pseudo-counts **are** the Beta posterior mean (Lecture 03) — Laplace smoothing (Lecture 03). "Rain always wets the grass" was an MLE on ten days.

Hidden variables leave no counts: infer them, re-count, repeat — EM, Lecture 07.

Learning the Structure

Score and search

Try graphs, score each by how well it fits the data *per parameter*, keep the best. The number of DAGs grows super-exponentially.

Test independences

Run conditional-independence tests and keep the graphs consistent with them — d-separation, read backwards.

Either way, data determines a graph only **up to Markov equivalence**: arrows between equivalent graphs are not in the data. A learned structure is not a causal discovery.

Mini-Exercise

Jack's lawn was wet on all **4** rainy days in a small sample.

What does counting give for $p(J{=}1 \mid R{=}1)$ — and what does one pseudo-count on each side give?

Mini-Exercise

Jack's lawn was wet on all **4** rainy days in a small sample.

What does counting give for $p(J{=}1 \mid R{=}1)$ — and what does one pseudo-count on each side give?

Counting: $4/4 = 1$ — "rain always wets the lawn", from four days. **Smoothed:** $5/6 \approx 0.83$, the posterior mean under $\mathrm{Beta}(1,1)$. Same data, two readings: the peak and the mean.

Takeaway

A graphical model makes the joint **writable**; its **shape** decides whether it is **queryable** — elimination is cheap on chains and trees and hopeless on a lattice. With complete data the parameters are counts: the MLE, or with pseudo-counts the Beta posterior mean. The structure is learnable only up to equivalence.

Appendix

Uncertain and Unreliable Evidence

Uncertain (soft) evidence

We are not sure of the observation itself: a distribution $\tilde{x}$ over its states. Sum over them:

$$p(y \mid \tilde{x}) = \sum_{x} p(y \mid x) p(x \mid \tilde{x})$$

We look at the lawn from the window without touching it: $p(T{=}1 \mid \tilde{T}) = 0.8$.

Unreliable evidence

We keep the report but **replace** its table by our own judgement of how reliable it is:

$$p(x \mid \mathrm{pa}(x)) \to \tilde{p}(x \mid \mathrm{pa}(x))$$

Jack says his lawn is wet, but he did not check.

Both are drawn with a **dashed** edge.